Ecological Indicators 158 (2024) 111424

Contents lists available at ScienceDirect

Ecological Indicators

journal homepage: www.elsevier.com/locate/ecolind

A systematic review of studies involving canopy layer urban heat island: Monitoring and associated factors

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ARTICLE INFO

Keywords:
Canopy layer urban heat island
Apparent temperature
Land cover/land use
Heat weave
Air pollution
Remote sensing

ABSTRACT

The urban thermal environment is closely related to the well-being of many city dwellers. Rich achievements have been obtained for canopy layer urban heat island (CLUHI) studies. Nevertheless, the monitoring and associated factors of CLUHI have not been systematically and timely reviewed. Therefore, this paper aimed to solve this issue to some extent by reviewing the fruitful research progress of CLUHI from the above-mentioned two aspects. The main findings were as follows. (1) Eight methods were adopted to obtain near-surface temperature data for CLUHI research, including four observation, two numerical modeling, remote sensing and data assimilation methods. (2) Air temperature was usually used rather than apparent temperature indices. Obvious differences existed between them, especially under humid hot or cold windy conditions. (3) The CLUHI intensity was generally defined as the differences in temperature between urban and suburban or rural stations or regions by population, land cover/land use, etc, or derived using regression analysis between temperature and impervious surface percentage. (4) The diurnal, monthly, seasonal or interannual variation of CLUHI has been generally analyzed in various global regions. (5) Six types of factors were analyzed, including meteorological conditions, air pollution, socioeconomic factors, urban underlying surface condition, inland or coastal type and landform conditions, and combined effects of some factors. (6) Five potential research directions were proposed, including improvement of data acquisition, sharing and use, considering apparent thermal indices, focus on the rarely studied regions, cities and scales, improving the calculation of CLUHI intensity, and attention to more associated factors and driving force analysis methods. This review can provide references for future CLUHI research and the construction of climate-resilient, livable, low-carbon cities.

1. Introduction

According to the United Nations Department of Economic and Social Affairs, the global urbanization rate will increase from 56 % in 2021 to 68 % in 2050 (Habitat, 2022). In the future, the growing 2.2 billion urban residents will mostly live in underdeveloped Africa and Asia (Habitat, 2022). These changes in land covers, land uses and humans' social and economic activity mode and intensity lead to the well-known urban heat island (UHI) phenomena, which means higher temperature exists in the urban regions than outskirts and villages. In general, there are four types of UHI, including the canopy layer UHI (CLUHI), surface UHI (SUHI), boundary layer UHI (BLUHI) and sub-surface UHI (SubUHI) (Qian et al., 2022; Wang, 2021; Zhou et al., 2019). The CLUHI refers to the difference in urban and rural air temperature that occurs from the ground to the roof of urban buildings. It is closely related to human welfare due to its direct or indirect influences on human comfort and health, energy consumption, urban flood and waterlogging, creature activities, air quality, urban infrastructure, financial loss, etc. The CLUHI intensities (CLUHIIs) could be 6–7 in Bangkok, Thailand (Marks and Connell, 2023), 7 °C in Singapore (Chow and Roth, 2006), 8 °C in

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https://doi.org/10.1016/j.ecolind.2023.111424
Received 9 October 2023; Received in revised form 26 November 2023; Accepted 11 December 2023
Available online 16 December 2023

Hania, Crete, Greece (Kolokotsa et al., 2009), 7–9 °C in Angul-Talcher region, India (Singh et al., 2023) and 11.3 °C in Hong Kong (Memon and Leung, 2010).

Much research on CLUHI has been carried out and obtained remarkable achievements. Three phases can be divided for previous studies in this research direction (Fig. 1), including the initial stage (1967–1990), slow acceleration period (1991–2010) and accelerated development phase (2011–now). Since 2000, the number of related articles has shown a "J-shaped" growth. Both the research articles and reviews were counted here. The search formula was "TS = ((‘urban heat island’) NOT (‘surface urban heat island’ or ‘boundary layer urban heat island’ or ‘boundary layer heat island’ or ‘sub-surface urban heat island’))". The details for the basic information of published papers were shown in the supplement materials, including the interannual variation in the number of published papers (Texts S1), the evolution of main journals (Texts S2 & Table S1), countries or regions (Texts S3 & Table S2), affiliations (Texts S4 & Table S3) and authors (Texts S5 & Table S4).

Previous studies have reviewed the canopy heat island in different areas from different perspectives. Several representative ones were as follows. Rizwan et al. (2008) reviewed the generation, determination and mitigation of atmosphere and surface UHI. Santamouris (2014) summarized the impacts of CLUHI on heating and cooling energy. Leal Filho et al. (2017) reviewed the CLUHIIs in some typical countries, strategies and tools to mitigate UHI. Wu and Ren (2019) did a bibliometric review of past trends and future prospects in UHI research from 1990 to 2017. Wang et al. (2021) and Leal Filho et al. (2021) reviewed the impacts of cool pavements and green infrastructure on CLUHI and SUHI, respectively. Rajagopal et al. (2023) reviewed the impact of environmental measures on UHI. Moreover, some authors summarized several aspects of UHI studies in a specified country or region, such as India (Veena et al., 2020), China (Tian et al., 2021), Singapore (Aflaki et al., 2017; Roth and Chow, 2012), Kuala Lumpur, Malaysia (Aflaki et al., 2017; Ramakreshnan et al., 2018) and Hong Kong (Aflaki et al., 2017). It is meaningful and necessary to summarize the rich research in CLUHI. Nevertheless, the existing reviews cannot fully reflect these fruitful achievements, especially from the aspects of monitoring and associated factors of CLUHI. Thus, this paper aimed to fill this gap to some extent by first reviewing the near-surface temperature data obtained by eight methods and several proposed human thermal indices for CLUHI monitoring (Fig. 2). Secondly, we summarized two calculation method of CLUHII and their diurnal, monthly, seasonal and interannual variation in various global regions. Then reviewed six types of associated factors, and finally pointed out five potential research directions.

2. Monitoring of canopy layer urban heat island

2.1. Eight methods to obtain the near-surface air temperature data used to monitor canopy layer urban heat island

2.1.1. Observation method

The near-surface (usually at the height of ~ 2 m) temperature can be observed using the meteorological stations, high-density meteorological measurement network, in-situ fixed observation points and moving transects (Fig. 2 & Table 1). Meteorological station observation was the most traditional and widely used method to study CLUHI (Ngarambe et al., 2021; Richard et al., 2021; Sachindra et al., 2016; Wang et al., 2022; Zhou et al., 2023). It can obtain high-quality and long-term archived values of multiple meteorological elements, which is extremely conducive to analyzing the long-term variation of CLUHI and the impacts of several meteorological elements on it. The disadvantage of this approach was that the number of meteorological sites was very small, meaning that they were spread quite thinly. Therefore, this method was only suitable for macroscopic scale research, and cannot accurately reflect the CLUHI situation in complex urban underlying surfaces. Moreover, these standard weather stations needed to follow strict standards, so the CLUHI intensity (CLUHII) they monitored was likely to be different from where people lived and worked.

In the last 20 years, some cities have built high-density meteorological measurement networks (Figure S1), including Tokyo, Taipei, Washington. DC, Helsinki, Hong Kong, London, Shanghai, Beijing, Shenzhen, Tainan, etc (Chen et al., 2019; Honjo et al., 2015; Lei et al., 2014; Yang et al., 2013). These networks usually consisted of 16–200 measurement points within the entire city, which have collected highly spatiotemporal dense data for several years (Honjo et al., 2015). Moreover, some scholars installed a.

certain number of observation points by themselves in the specific small (Howe et al., 2017; Kadhim-Abid et al., 2019; Sfîcă et al., 2018; Shi et al., 2021; Tong et al., 2017) or entire urban central area (Xu et al.,

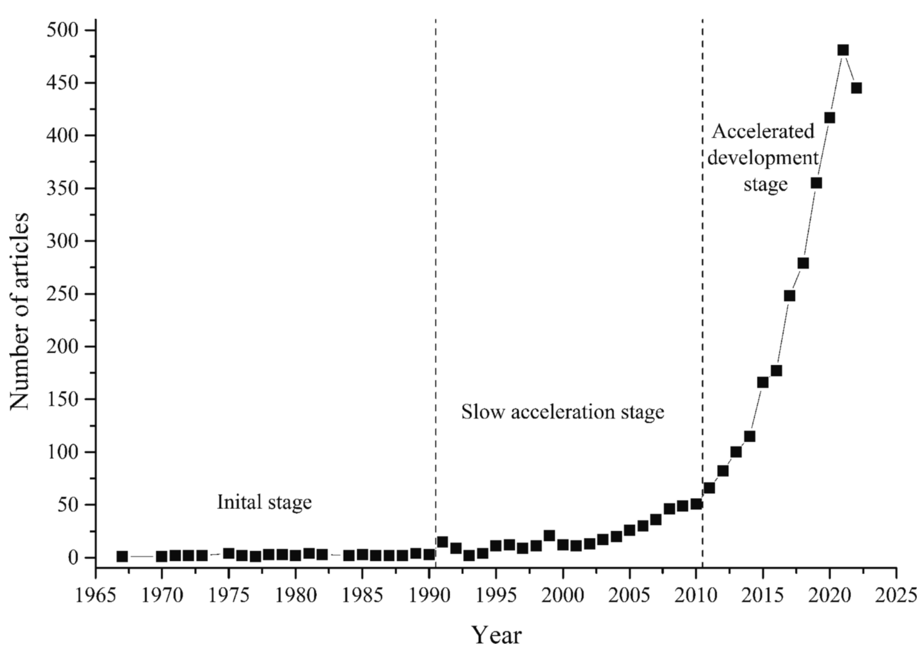

Fig. 1. The number of articles from 1967 to 2022 on canopy layer urban heat islands.

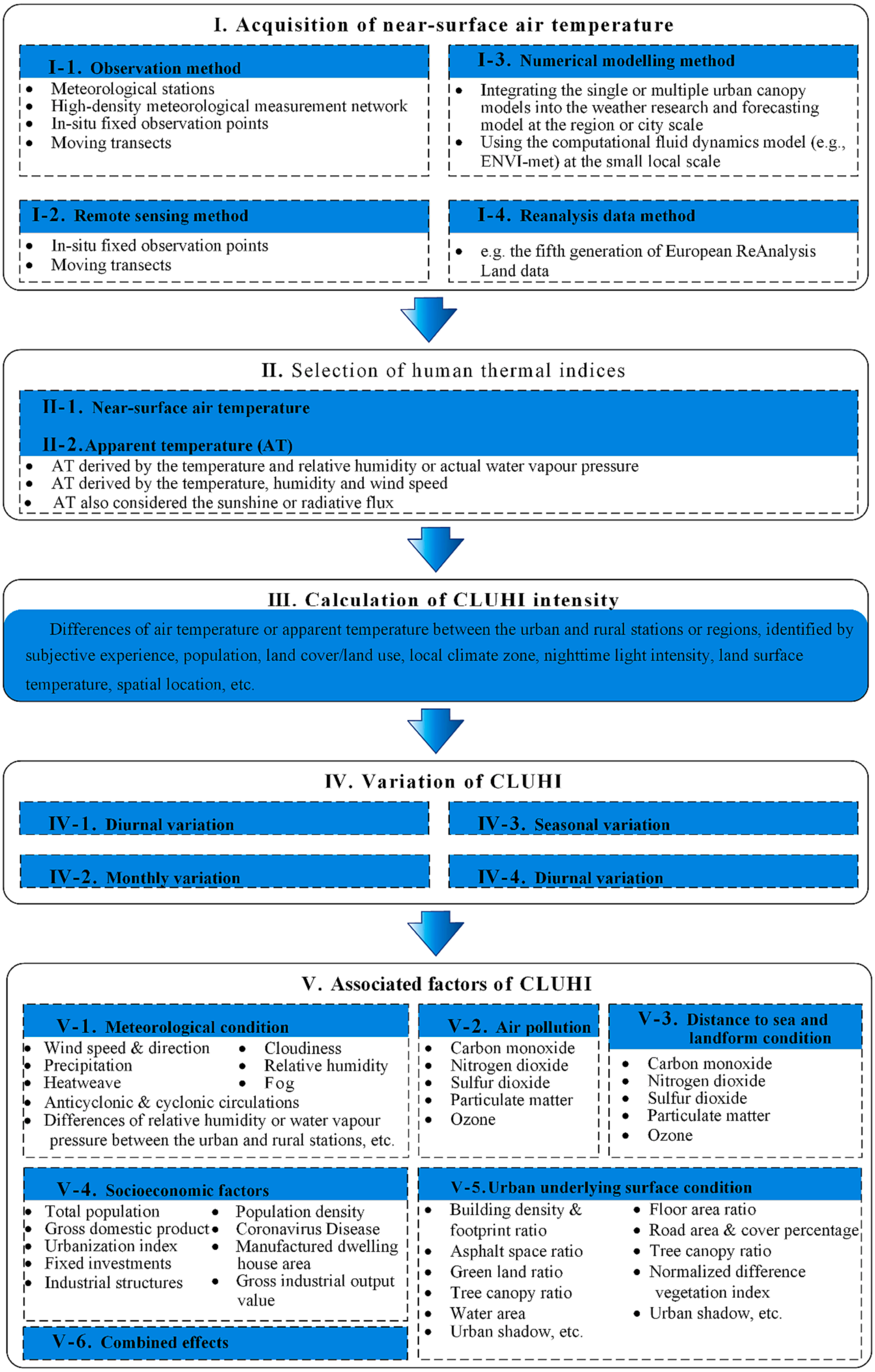

Fig. 2. Research framework on monitoring and associated factors of canopy layer urban heat islands in previous studies.

Table 1
Eight methods to acquire the near-surface air temperature data used to study canopy layer urban heat islands.

Technological means	Specific method	Data accuracy	Spatial range	Spatial accuracy	Time range	Time resolution	Parameters except temperature	Representative studies
Observation	Meteorological station observation	Quite high, but the temperature may not be the same as in the urban areas where people work and live.	Large scale or even the world	Sparse sites in large areas	Several decades	Hourly, multi-hourly, daily, monthly, yearly	Several indices	Ren et al. (2021); Rogers et al. (2019); Wang et al. (2022)
	High-density meteorological measurement network	Accuracy may be affected by poor ventilation or radiation due to the limited observation conditions in the complex urban environment.	Entire city.	Usually 16–200 measurement points in the entire city	Several years	≥10 min	A few or several indices	Chen et al. (2019); Honjo (2019); Richard et al. (2021)
	In-situ fixed-points observation	Accuracy may be affected by poor ventilation or radiation due to the limited observation conditions in the complex urban environment.	The local region inside the urban area	A certain number of sites in a small region	Several days, months or years	Hourly	None or a few indices (relative humidity, wind speed, wind direction, air pressure)	Howe et al. (2017); Sfîcă et al. (2018); Shi et al. (2021)
	Moving transect observation	Accuracy may be affected by the hysteresis effect of the probe, passing vehicles and the observing vehicle itself, external radiation, synchronization correction, etc.	Specified route(s) in a small region	Specified route (s) in a small region	Most one-day or several days	As required	Relative humidity	Chàfer et al. (2022); Chen et al. (2023); Shi et al. (2021)
Remote sensing	Remote sensing	Accuracy is difficult to evaluate in the complex urban environment and the retrieved temperature is not as good as that observed, especially under cloudy conditions.	Large scale or even the world	60–120 m for Landsat thermal data, 1 km for Terra or Aqua data, etc.	Most one-day, one-year or several decades	Specified date, hourly, monthly, or seasonally	None	Du et al. (2023); Ho et al. (2016); Yao et al. (2021)
Numerical modeling	The urban boundary layer model coupled with the urban canopy model	Accuracy is difficult to evaluate in the complex urban environment, and the accuracy can be affected due to the quality of input data, limitations and uncertainty of the simulated model.	Large scale (usually the entire city or region with a large area)	$\geq$ 100 m, but usually $\geq$ 1 km	Several days	Hourly	Several indices	Silva et al. (2022); Singh et al. (2023); Wang and Li (2021)
	Computational fluid dynamics	Accuracy can be affected due to the quality of input data, limitations and uncertainty of the simulated model.	Small scale (usually building block or region with a small area)	0.5–10 m	One or two days	$\geq$ 10 s	Several indices	Faragallah and Ragheb (2022); Najah et al. (2023); Zheng et al. (2022)
Data assimilation	Reanalysis data	Accuracy is difficult to evaluate in the complex urban environment, and the accuracy can be affected due to the quality of input data, limitations and uncertainty of assimilated and simulated models.	Large scale or even the world	$\geq 0.1^\circ$	Several decades	Hourly	Several indices	Das et al. (2020); Lee et al. (2019)

2019). These data have provided great convenience for analyzing the spatiotemporal changes in the thermal environment in complex cities (Chapman et al., 2017; Chen et al., 2019; Honjo, 2019; Honjo et al., 2015; Richard et al., 2021). Some issues still existed for these studies. Firstly, the highly spatiotemporal dense data were not easy to obtain in many cases. Secondly, these sites may not be subject to strict observation conditions, and the data accuracy may be affected by poor ventilation or radiation due to the limited observation conditions in the complex urban environment. Thirdly, these networks usually did not monitor as many indices as the traditional weather stations, or even just observe temperature, especially for stations built by the authors themselves mainly due to the limited research funding. Fourthly, the installation of observation facilities was likely to be opposed by managers, and even the equipment can be easily damaged or stolen by some people. Finally, the data archived by this method was not long enough, especially for the self-constructed observation points. This made it impossible to study the long-term trend of CLUHI.

The moving transect observation method installed the monitoring instruments in the moving cars, trams, electric mopeds or bicycles, and obtained dense temperature distribution data over one or several given routes in a short time in a specified area (Figure S2) (Chàfer et al., 2022; Chen et al., 2023; Liu et al., 2017; Oliveira et al., 2011; Shi et al., 2021; Shumilov et al., 2017; Zheng et al., 2023b). It was beneficial to describe the thermal environment within a complex city. Some disadvantages were as follows. Firstly, these observed temperature data needed to be calibrated to the same time based on the temperature change laws at the reference sites, to ensure that they could be comparable to each other (Chen et al., 2023; Liu et al., 2017; Zhou and Chen, 2021). The assumption was that the temperature variation law at each observation point in the routes was consistent with the reference points. However, there were significant differences in the urban temperature changes under different biophysical conditions. Therefore, the observation time was inappropriate too long and the research area should not be too large for this kind of research. Secondly, the data accuracy may be affected by the hysteresis effect of the probe, passing vehicles or the observing vehicle itself, external radiation, synchronization correction, etc. The hysteresis effect of the probe in this paper meant a certain dwell time should be spent for the probe to accurately monitor the temperature at a given point, but the mobile device kept moving. This could lead to some bias or even errors, especially for the mobile device at high speed. Thirdly, this method mainly monitored temperatures on roads, but it did not work well for other parts of the city. Finally, the equipment was generally not suitable for rainy weather monitoring.

2.1.2. *Remote sensing method*

The remote sensing method aimed to analyze the differences between the urban and rural areas for the near-surface air temperature or apparent temperature retrieved by satellite images (Table 1, Fig. 3& S3) (Anniballe et al., 2014; Du et al., 2023; Ho et al., 2016; Huang et al., 2017; Yao et al., 2021; Zou et al., 2021). The spatial resolution of those derived air temperature indices was about 100 m (Ho et al., 2016) or 1 km (Du et al., 2023; Huang et al., 2017; Li and Zha, 2019; Yao et al., 2021). They can be the momentary values (such as at 14:00) (Zou et al., 2021), maximum (Ho et al., 2016) or hourly values (Li et al., 2017) on a specified date, monthly (Huang et al., 2017), monthly mean values at four moments during the satellite's transit (Huang et al., 2017), seasonally mean values at 14:00 and 02:00 (Li and Zha, 2019), and seasonally mean, maximum and minimum values (Yao et al., 2021). These near-surface air temperatures or apparent temperatures were usually derived using the land surface temperature (LST), atmospheric water vapor, and other auxiliary influence factors, such as remoted sensed LC/LU indices, topographic factors, nighttime light intensity (NLI), solar radiation, sky view factor, distance from the ocean, etc. (Ho et al., 2016; Huang et al., 2017; Yao et al., 2021; Zou et al., 2021). The retrieval models included the Cubist matching learning algorithm (Yao et al., 2021), the random forest regression model (Ho et al., 2016; Li et al., 2017; Li and Zha, 2019; Zou et al., 2021), multiple linear regression analysis (Huang et al., 2017), etc.

This method was extremely beneficial in analyzing the detailed characteristics of CLUHI within the urban areas. Nevertheless, the validity of this method for CLUHI analysis deserves further study. On the one hand, the relationship was complicated between the near-surface temperature indices and LSTs, especially in the urban areas with strong heterogeneity, although there were close relationships between them. On the other hand, it was difficult to evaluate the accuracy of the derived temperature indices by remote sensing in the complex urban context with highly mixed pixels. Most of the root-mean-square errors (RMSEs) were between 1 and 3 °C in previous studies (Zou et al., 2021), which may not be effective in quantifying CLUHI, especially when the CLUHII was not high. Previous studies have compared the differences between the CLUHI and SUHI using the derived near-surface air temperature, apparent temperature and LSTs (Ho et al., 2016; Li and Zha, 2019; Yao et al., 2021). However, to our knowledge, there were rare reports on the analysis of the differences in CLUHII between the temperature observed by weather stations and those retrieved by remote sensing by far. Finally, this method was not recommended under cloudy or rainy weather condition due to the large uncertainty for these retrieved temperature indices. They were usually derived using the observed meteorological data under bad weather, and remote sensed parameters before or after bad weather.

2.1.3. *Numerical modelling method*

The numerical modeling method has been widely used for CLUHI studies (Table 1). They mainly included integrating the single or multiple urban canopy models (UCM) into the weather research and forecasting (WRF) model at the region or city scale (Silva et al., 2022; Singh et al., 2023; Wang and Li, 2021), and using the computational fluid dynamics model (CFD) (e.g., ENVI-met) at the small local scale (Faragallah and Ragheb, 2022; Najah et al., 2023; Zheng et al., 2022). The spatial resolution of the WRF/UCM model was usually no less than 1 km but could reach 100 m. Meanwhile, the resolution ranged from 0.5 m to 10 m for the CFD model. A great advantage of the numerical simulation method was that it could not only simulate the characteristics of the CLUHI in the current study area but also predict and optimize the future situation under different planning or climate change scenarios (Faragallah and Ragheb, 2022; Silva et al., 2022; Zheng et al., 2022).

The WRF/UCM could reproduce the built-up area environment with the generalized urban structure and geometrical configuration (Khan

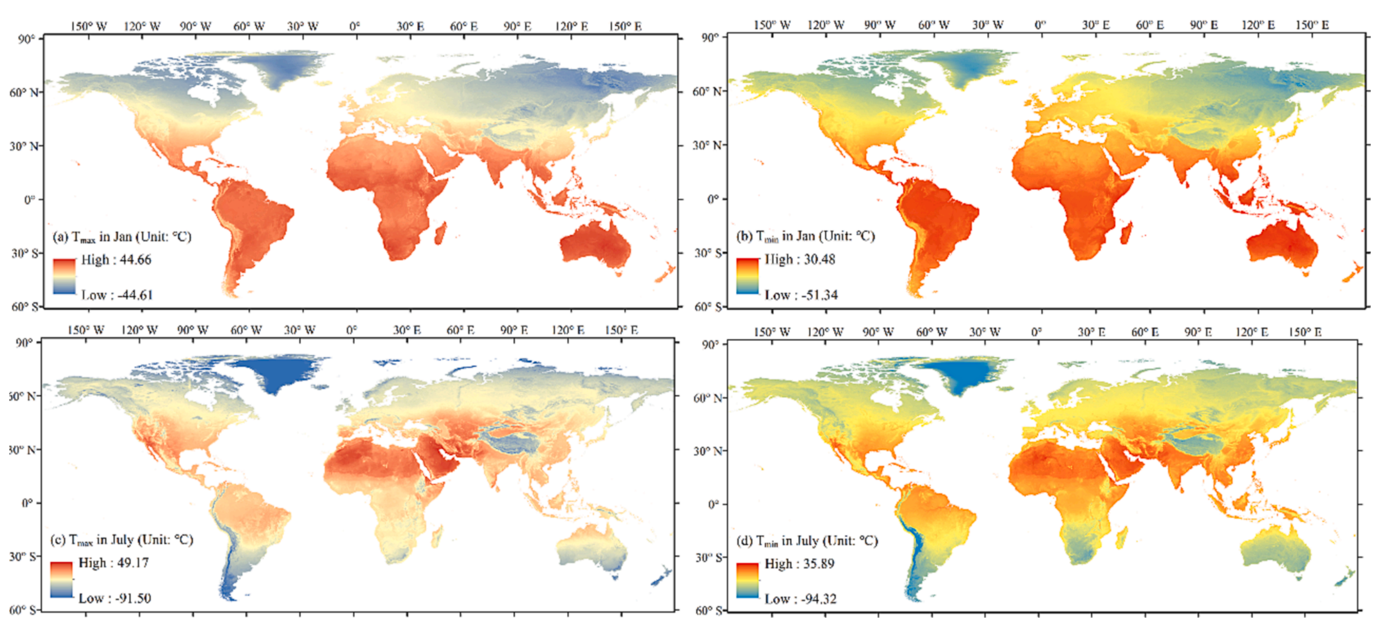

Fig. 3. Spatial distributions of the remotely sensed maximum (Tmax) (a) and minimum air temperature (Tmin) (b) in January, Tmax (c) and Tmin (d) in July during 2016–2020 in mainland China. Data source: Middle Yangtze River Geoscience Data Center (https://cjgeodata.cug.edu.cn/#/pageDetail?id=97).

et al., 2021). It could capture the shadow areas of buildings, short and long radiations, wind flow through the canopy, and multilayer heat transfer for building roofs, concrete walls and road pavement surfaces, etc. (Khan et al., 2021). Nevertheless, limited by the spatial resolution of the model and its defects, the WRF/UCM method was unable to simulate many urban details and ecological processes, such as the uneven heights of urban buildings, vegetation activity, hydrological process, etc. The parameterization scheme was complicated for the physical processes in the model. To obtain high-precision simulation results, the researchers needed to master solid knowledge of meteorology and provide a large amount of background data with high accuracy in the study area.

ENVI-met was the most widely used CFD model, whose biggest advantage was the consideration of the effects of vegetation and waterbody on meteorology (Najah et al., 2023; Xu et al., 2022; Zheng et al., 2022). High-precision results could be obtained for many meteorological elements in three dimensions. ENVI-met was especially suitable for the study of the influence mechanism, optimizing strategy and planning schemes on the micrometeorology at the scale of single buildings, building groups and street blocks. The ENVI-met model also required many input parameters (Table S5), whose accuracy could greatly affect the quality of simulation results. Moreover, the ENVI-met model was only suitable for the clear sky condition. In addition, it required a lot of computing time and good computer performance, and could only simulate a very limited area. Finally, the applicability of ENVI-met needed further verification in different global climatic zones.

2.1.4. Reanalysis data method

The reanalysis data was a set of grid data for many long-time meteorological indices (Hersbach et al., 2020; Hoffmann et al., 2019; Zou et al., 2022). These data were derived by reprocessing and analyzing historical meteorological observation data using the advanced numerical prediction model and assimilation analysis technique. The representative data was the fifth generation of European ReAnalysis Land data (ERA5-Land). It has the hourly 0.1° resolution near-surface (2 m height) air temperature and other 49 meteorological indices, covering most parts of the globe from 1959 to the present. Compared with the observed values in meteorological stations, the RMSEs were 1.85, 1.90 and 1.91 °C for the simulated hourly temperature in the croplands, forests, urban and built-up lands in the Guangdong – Hong Kong – Macau Greater Bay Area, China, respectively (Zou et al., 2022). Meanwhile, the RMSEs turned out to be smaller for the daily (1.38, 1.44 and 1.50 °C) and monthly values (1.11, 1.18 and 1.24 °C) in these land covers (Zou et al., 2022).

Das et al. (2020) used the 0.125° resolution ERA-Interim data to explore the CLUHI in the summer in Kolkata, India. Lee et al. (2019) analyzed the CLUHI for eight megacities in East, South and Southeast Asia, using the monthly downscaled and assimilated mean monthly temperatures at 1 km resolution. Similar studies rarely existed and should be encouraged for large and megacities. Some issues or uncurtains existed. On the one hand, the spatial resolution was too coarse for CLUHI study in the vast majority of cities due to limited urban areas. On the other hand, the accuracy was too difficult to evaluate in the complex urban environment for these reanalysis data. Some drawbacks existed regarding the reanalysis data, including the neglect of clouds and surface moisture, changes in the atmospheric composition effects (Trenberth, 2004), non-homogenization and time-varying of the input data (Parker, 2010), bias associated with the income and expense balance of surface heat (Trenberth, 2004), etc.

2.2. Human thermal indices for canopy layer urban heat island studies

Much of the existing research used the near-surface air temperature to study CLUHI (Fig. 2, Tables S6 & S7). Although temperature was the most important factor affecting thermal comfort, other meteorological factors could not be neglected, including relative humidity, wind speed, solar radiation, etc. (Luo and Lau, 2019; Zhang et al., 2023).

Several human thermal indices were derived by the temperature and relative humidity or actual water vapor pressure, including the indoor apparent temperature (Steadman, 1979), the heat index (Rothfusz and Headquarters, 1990), Humidex (Masterton and Richardson, 1979), discomfort index (Sohar et al., 1963), etc. Some indices were derived from the temperature, humidity and wind speed, such as outdoor apparent temperature (Steadman, 1984), net effective temperature (Yaglou, 1923), wind chill temperature (Osczevski and Bluestein, 2005), etc. Some indices also considered the sunshine or radiative flux, including the wet bulb globe temperature (Yaglou and Minaed, 1957), predicted mean vote (Fanger, 1984), universal thermal climate index (Jendritzky et al., 2012), physiological equivalent temperature (PET) (Höppe and Mayer, 1987), etc.

Obvious differences exist between the air temperature and apparent temperature, especially under hot humid or cold windy weather conditions (Luo and Lau, 2019). Nevertheless, rare studies analyzed CLUHI by taking the temperature and other meteorological factors into account. Basara et al. (2010) compared the differences of two apparent indices between the urban or suburban and rural sites during an intense heatwave in Oklahoma, United States. Katavoutas et al. (2015) analyzed the three hourly differences in PET between the urban and rural stations during the summer, winter and transitional periods. Ho et al. (2016) used the retrieved maximum daily apparent temperature by remote sensing to analyze the CLUHI in the greater Vancouver area, Canada. Harmay and Choi (2023) analyzed the differences in both the air temperature and heat index between the urban and rural pixels in four selected cities over the conterminous United States based on climate zones and land cover classifications.

2.3. Calculation of canopy layer urban heat island intensity

The CLUHII was generally defined as the differences in near-surface temperature between the urban and suburban or rural stations or regions (Fig. 2 & Table S7). The classification of weather stations was according to various criteria, including the subjective experience (Levermore et al., 2018; Lokoshchenko, 2017; Wang et al., 2022), population (Rogers et al., 2019; Scott et al., 2018; Stone Jr, 2007), land cover/land use (LC/LU) (Chow and Roth, 2006; Sun et al., 2022; Zhou et al., 2023), local climate zone (LCZ) (Basara et al., 2010; Golroudbary et al., 2018; Richard et al., 2021), NLI (Stone Jr, 2007), LST (Jia et al., 2019; Yang et al., 2019; Yang et al., 2013), spatial location (Yang et al., 2019; Yang et al., 2013), etc.

Some representative studies were as follows. Sun et al. (2022) defined the urban stations as those with more than 70 % percentage of IS (PIS) within the 1–5 km buffer zones, while the rural stations as those with less than 3 % and 15 % PIS within the 1–5 and 7–78 km buffer zones, respectively. Moreover, the stations should be more than 1 km away from the waterbody at 30 m resolution. The absolute differences in the average elevation between the urban and rural stations should be less than 30 m. Zhou et al. (2023) delineated the urban borders by clustering the land parcels with the PIS equal to and larger than 50 %. The reference areas referred to 50 km buffer zones near the urban boundaries. Urban stations were also defined as those in compact mid-rise LCZ, while rural were in sparsely built or unbuilt LCZs (Richard et al., 2021). The rural station in Singapore referred to was in a large less less-developed area mixed with secondary tropical rainforest, cropland and military training grounds (Chow and Roth, 2006).

Moreover, Rogers et al. (2019) mainly defined the urban, suburban and rural stations as those within a certain population range in the urban areas, 2 km buffer zones or outside regions. Nevertheless, there was no uniform standard on how to determine the reasonable population threshold and statistical space extent. Different countries and regions have various population standards for cities. Further, Stone Jr (2007) pointed out that the distribution of population may not always correlate highly with the location of surface infrastructure, such as in the case of a rural airport or military installation. Then he used not only the total population but also NLI to classify stations. Scott et al. (2018) defined urban stations as those with the highest satellite nighttime light intensities in each metropolitan area with more than 500,000 urban populations in the United States. Meanwhile, the rural stations were defined as those with the lowest nighttime intensities within 30 km of their corresponding urban stations. The urban and reference stations could also be determined based on the contour maps of LST (Jia et al., 2019; Yang et al., 2019; Yang et al., 2013). The identification of rural sites also considered other constraints (Jia et al., 2019). For instance, the distance to the boundaries of built-up areas should be within 30 km to ensure the rural sites were in the same large-scale circulation and climate background as the corresponding urban stations. The absolute values of elevation differences between the urban and rural stations should be less than 20 m, avoiding the disturbance brought by height differences. In addition, the urban stations were defined as those located within the sixth ring in Beijing (Yang et al., 2019; Yang et al., 2013). Finally, some scholars avoided disputes over the definition of urban and rural areas and instead defined the CLUHII as the temperature differences between regions with 0 % and 100 % PIS based on the regression analysis between temperature and PIS (Cecilia et al., 2023; Schatz and Kucharik, 2015).

2.4. Spatiotemporal variation rules of canopy layer urban heat island

2.4.1. Diurnal variation

The CLUHIIs were generally larger during the nighttime than daytime (Al-Saadi et al., 2020; Golroudbary et al., 2018; Guattari et al., 2018; Li et al., 2018a; Ramakreshnan et al., 2018) except in Hania, Crete, Greece (Kolokotsa et al., 2009) (Fig. 2 & Table S7). During the summer months, the CLUHII showed a hump-shaped diurnal variation pattern in most American cities (Ramamurthy and Sangobanwo, 2016). The CLUHII was higher during the nighttime and then dropped steadily after sunrise (Ramamurthy and Sangobanwo, 2016). One exception was in Seattle, where the CLUHII decreased during the nighttime while increasing during the daytime (Ramamurthy and Sangobanwo, 2016). In the winter month, the daily CLUHI changed subtly and showed a near-flat profile in most American cities (Ramamurthy and Sangobanwo, 2016).

The common diurnal variation pattern contained two relatively stable stages separated by two swift changing stages in Trento, Italy (Giovannini et al., 2011), Rome, Italy (Cecilia et al., 2023), Londrina, Brazil (Targino et al., 2014), Seoul, Korea (Park et al., 2023), Beijing (Yang et al., 2019) (Figure S4) and Xi'an, China (Xu et al., 2019). The CLUHII was usually weaker around noon and stronger at night. It declined rapidly in the early morning and increased significantly in the late afternoon and evening. Besides, the CUUHII was usually strongest at ~ 0:00, then declined steadily before sunrise. This change pattern was also suitable for Iași, Romania in all seasons except for winter (Sfîcă et al., 2018). In addition, the CLUHII was stronger during the nighttime while weakest at 16:00 in Granada, Spain (Montávez et al., 2000).

Moreover, the CLUHII was the weakest during the midday phase, increased continually until midnight, strongest after midnight and declined sharply after sunrise in Bilbao, Spain (Acero et al., 2013), Ulaanbaatar, Mongolia (Ganbat et al., 2013) and Hong Kong, China (Memon and Leung, 2010). The CLUHII decreased in the morning, then stayed weak through the midday phase, increased to the peak, and then decreased before midnight in the hot–humid tropical Akure, Nigeria (Balogun et al., 2010). The CLUHII was stronger during the night and weaker during the morning and with a rapid decline after sunrise in Agrinio, Greece (Vardoulakis et al., 2013). The CLUHII was stable at high values overnight and the lowest at noon in Singapore (Chow and Roth, 2006). The diurnal change pattern presented an approximate sinusoidal curve, stronger at night, weaker in the morning, strongest at around 18o'clock, and weakest at 8–9o'clock in the Netherlands (Golroudbary et al., 2018; Wolters and Brandsma, 2012). The longer the number of continuous precipitation hours during the daytime, the more this classic pattern changed (Yang et al., 2019). Similarly, the diurnal pattern of CLUHII changed and no obvious variations existed for the diurnal CLUHI under the cold front (Targino et al., 2014). In addition, the diurnal pattern of CLUHII changed significantly when using the PET rather than air temperature (Fig. 4) (Katavoutas et al., 2015). Its three-hourly interval curve showed the shape of a sinusoidal curve. The CLUHII was strongest at noon, weakest in the morning, and stronger before midnight than after midnight in Athens, Greece (Katavoutas et al., 2015).

2.4.2. Monthly or seasonal variation

Previous studies have analyzed both the monthly (Founda et al., 2015; Ngarambe et al., 2021; Schlünzen et al., 2010) and seasonal variation of CLUHII (Peng et al., 2019; Sfîcă et al., 2018; Yang et al., 2021). The season's order was summer > spring > autumn > winter for all-day mean CLUHII of 155 cities in China (Peng et al., 2019), Ulaanbaatar, Mongolia (Ganbat et al., 2013) and Netherlands (Golroudbary et al., 2018; Wolters and Brandsma, 2012). The CLUHII was stronger in the summer than in winter in Poznań, Poland (Półrolniczak et al., 2017), Eilat, Israel (Sofer and Potchter, 2006), Tianjin (Tong et al., 2017), Wuhan (Jia et al., 2019), China. The CLUHII derived by PET was also stronger in the summer than in winter and transitional period in Athens, Greece (Katavoutas et al., 2015). In addition, Sfîcă et al. (2018) found the diurnal range of CLUHII was the largest in the summer, middle in the spring and autumn, and smallest in the winter. Both the strongest and weakest CLUHII occurred in the summer (Scott et al., 2018). Moreover, the season order of CLUHII was autumn > winter > spring > summer in Soul, Korea (Kim and Baik, 2002), autumn > spring > winter > summer in Shanghai, China (Zhang et al., 2010), and autumn > winter > summer > spring in Bilbao, Spain (Acero et al., 2013). The month order of CLUHII was September > July > August > October > June in Hania, Crete, Greece (Kolokotsa et al., 2009).

Larger CLUHII generally occurred in the winter than in summer in 86 cities (Zhou et al., 2023), Beijing (Yang et al., 2013; Yu, 2005), Nanjing (Zeng et al., 2009) in China, Agrinio in Greece (Vardoulakis et al., 2013). The same rules also existed in most years rather than all years in Nicosia, Cyprus (Theophilou and Serghides, 2015). The season order was winter > autumn > summer > spring for CLUHII in Granada, Spain (Montávez et al., 2000). The season order of CLUHII was winter > summer > rainy season in the coastal region in Pakistan (Rizvi et al., 2019). The CLUHII was slightly stronger during dry months in Trento, Italy (Giovannini et al., 2011). The CLUHII was lower during the Northeast monsoon period and slightly higher during the Southwest monsoon stage in Singapore (Chow and Roth, 2006). The CLUHII was larger in December and November, while weakest in April in Hong Kong, China (Memon and Leung, 2010). Only negative CLUHII only occurred in November in Cairo, Egypt (Robaa, 2003).

The CLUHII derived by the minimum temperature showed greater seasonal variation than the maximum temperature in Salamanca, Spanish (Alonso et al., 2003), Baghdad, Iraq (Al-Saadi et al., 2020) and Beijing, China (Yang et al., 2021). The opposite finding existed in Athens, Greece (Founda et al., 2015). Meanwhile, the CLUHII showed fewer monthly or seasonal changes during the nighttime than daytime in Hamburg, German (Schlünzen et al., 2010). Moreover, the CLUHII derived by the maximum and minimum temperature showed similar seasonal variation amplitude but different seasonal change order in Barcelona, Spanish (Moreno-garcia, 1994).

The nighttime CLUHII was larger in the summer half-year in Hamburg, Germany (Schlünzen et al., 2010). Similarly, the CLUHII derived by the minimum temperature was strongest in the summer in Baghdad, Iraq (Al-Saadi et al., 2020). However, the CLUHII derived by the minimum temperature was larger in the autumn than in other seasons in Granada, Spain (Montávez et al., 2000). Similarly, the nighttime CLUHII was larger in the spring and autumn than in the winter and summer in Tainan, China (Chen et al., 2019). Moreover, the season order of CLUHII derived by the minimum temperature was dry season > hot

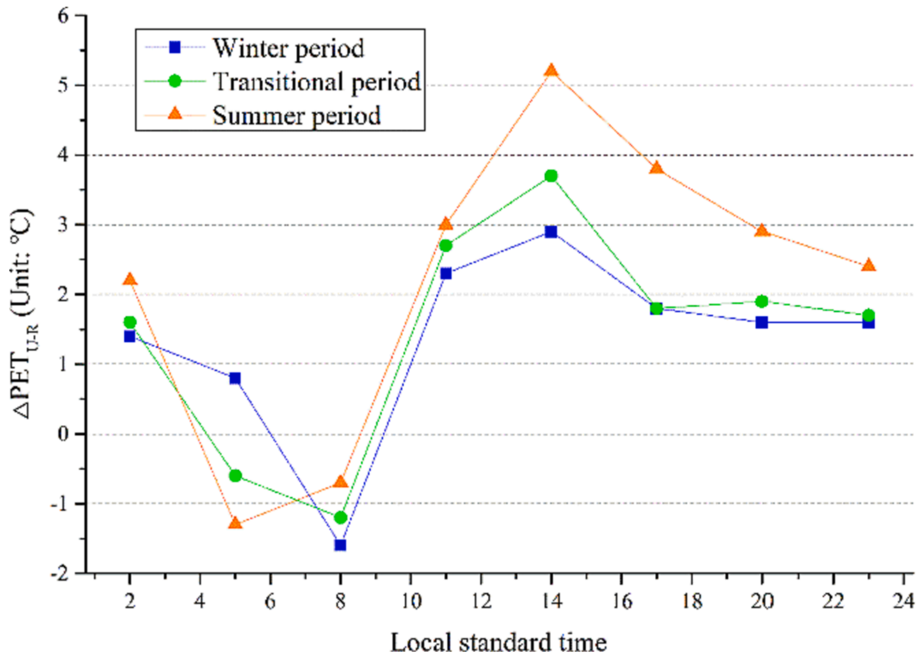

Fig. 4. Differences in the physiological equivalent temperature between the urban and rural sites (ΔPET_{u-r}) for every three hours in Athens, Greece. Data was obtained by Katavoutas et al. (2015).

season > wet season in Khartoum, Sudan (Elagib, 2011). However, the nighttime CLUHII showed complex variation rules in Athens, Greece (Founda et al., 2015).

The daytime CLUHII was smallest in the winter, while largest in the summer in Athens, Greece (Founda et al., 2015), and least in the winter while largest in the spring in Tainan, China (Chen et al., 2019). Meanwhile, the CLUHII derived by the maximum temperature was the strongest in the summer and weakest in the autumn in Barcelona, Spain (Moreno-garcia, 1994). However, the season order was winter > autumn > summer > spring in Granada, Spain (Montávez et al., 2000). Moreover, the CLUHII derived by the maximum temperature was slightly higher in the dry season than in the hot and wet seasons in Khartoum, Sudan (Elagib, 2011). In addition, it showed a small seasonal variation in Salamanca, Spanish (Alonso et al., 2003), and no obvious seasonal change existed in Baghdad, Iraq (Al-Saadi et al., 2020) and Beijing, China (Yang et al., 2021). Nevertheless, the previous studies generally just compared the CLUHIIs simply to study the monthly or seasonal variation. The statistical test has been rarely done on whether there were significant differences in CLUHII in various months or seasons.

2.4.3. Interannual variation

The CLUHII usually increased continuously, but the rate varied during the daytime and nighttime in different seasons and regions under various contexts. Nevertheless, CLUHII changed little (Zhang et al., 2022a) or even decreased in some cases (Jia et al., 2019; Theophilou and Serghides, 2015). Some representative studies were as follows. Based on the remoted sensed temperature data, the increase magnitudes were all 0.03 ± 0.002 °C/decade for the daytime and nighttime CLUHII in global 5643 cities during 2003–2020 ($p < 0.05$) (Du et al., 2023). The mean decadal change rate was 0.05 °C in the CLUHII of 50 large American cities from 1951 to 2000 (Stone Jr, 2007). The annual mean CLUHII increased significantly from 1984 to 2013 for 155 cities in China ($p < 0.001$) (Peng et al., 2019). The CLUHII exhibited a clear interannual warming trend with a rate of 0.31 °C/10a during 1960–2000 in Beijing, China (Yu, 2005), 0.21 °C/10a during 1996–2011 in Manchester, United Kingdom (Levermore et al., 2018) and 0.08 °C/a during 2008–2019 in Baghdad, Iraq (Wahab et al., 2022). The annual mean CLUHII was nearly 1.0 °C at the end of the 19th century, 1.2 °C one century ago, 1.5–1.6 °C in the middle and at the end of the 20th century, and 2.0 °C in recent years (2010–2014) in Moscow, Russia (Lokoshchenko, 2017). Both the CLUHII derived using the minimum and maximum temperature increased from 2008 to 2019 in Baghdad, Iraq (Al-Saadi et al., 2020). Comparing data in 1990 and 2005, the CLUHII increased in most months, but to different degrees in Hong Kong, China (Memon and Leung, 2010). The all-day and daytime CLUHII showed a gradual and constant increase during 1970–2004, while the nighttime CLUHII increased only in the summer after the mid-1980 s in Athens, Greece (Founda et al., 2015).

The CLUHII derived by the minimum and mean temperatures showed a significant increase, while no significant enlargement was found for CLUHII derived by the maximum temperature in Chongqing, China (Zhang et al., 2022a). The annual mean CLUHII showed a clear rising trend during the nighttime unlike the daytime from 1952 to 2010 (Sachindra et al., 2016). Moreover, the CLUHII derived by the minimum temperature would be larger in all seasons in the future based on the predicted results by the downscaling general-circulation model (Sachindra et al., 2016). For the period 1961–2004, CLUHII underwent a significant increase, especially during 1988–2004 (Jia et al., 2019). However, a significant decrease existed during 2004–2015, and the change amplitude was larger for CLUHII derived by the minimum than maximum temperature (Jia et al., 2019). In addition, the CLUHII continued to decrease in the three decades from 1983 to 2010 in Nicosia, Cyprus (Theophilou and Serghides, 2015).

3. Associated factors of canopy layer urban heat island

3.1. Meteorological condition

The CLUHII was usually positively correlated with the CLUHII in the previous day (Kim and Baik, 2002; Kim and Baik, 2004; Zheng et al., 2023a), sea level air pressure (Wolters and Brandsma, 2012) and the differences in water vapor pressure (Robaa, 2003) (Fig. 2 & Table S7). Meanwhile, negative relationships often existed between CLUHII and wind speed (Ngarambe et al., 2021; Peng et al., 2019; Schlünzen et al., 2010; Wolters and Brandsma, 2012; Zheng et al., 2023a), cloudiness

(Ngarambe et al., 2021; Park et al., 2023; Sundborg, 1950; Wolters and Brandsma, 2012; Zheng et al., 2023a), precipitation (Kolokotsa et al., 2009; Peng et al., 2019), relative humidity (Kim and Baik, 2004; Park et al., 2023; Peng et al., 2019; Zheng et al., 2023a) and the differences of relative humidity between the urban and rural stations (Robaa, 2003). However, Rogers et al. (2019) found no significant pattern in the effect of sea breeze on the CLUHI in three coastal cities in Southeast Australia. Acero et al. (2013) revealed the effects of wind speed on CLUHII depended on the wind direction in the coastal Bilbao city in northern Spain. Obvious negative impacts existed when the airflow was from the sea, but little effect when the airflow was towards the sea. Kolokotsa et al. (2009) found the northern winds could expand the CLUHI front, while the western winds reduced CLUHII in Hania, Crete, Greece. Moreover, the negative impacts of cloudiness on CLUHII possibly existed, but could not be substantiated in Hong Kong, China (Memon and Leung, 2010). In addition, the CLUHII did not necessarily enlarge under stable conditions with low wind velocity due to the accumulation of cold air in the valley, especially in the early morning hours in Eilat, Israel (Sofer and Potchter, 2006).

Fog decreased CLUHII by about 1 °C under cloudless and windless weather conditions during the anticyclonic synoptic situation in the valley floor areas in Kraków, Poland (Bokwa et al., 2018). The CLUHII was higher in the anticyclonic than cyclonic circulations in Poznań, Poland (Półrolniczak et al., 2017), and under stable than unstable weather conditions in Salamanca, Spanish (Alonso et al., 2003). The CLUHII was largest (6.6 °C) under lingering high-pressure conditions, milder (3.0 °C) under cold anticyclones and almost vanished (1.0 °C) during the passage of cold fronts in Londrina, Brazil (Targino et al., 2014).

Sundborg (1950) proposed the first empirical relationship between CLUHII and two meteorological factors (cloudiness and windspeed) in Uppsala, Sweden. Kim and Baik (2002) used the multiple linear regression method and artificial neural network model to construct the relationship between CLUHII and four influence factors in Seoul, Korea, including the wind speed, cloudiness, relative humidity and previous-day maximum CLUHII. Levermore and Parkinson (2017) built the multiple linear regression model between CLUHII and wind speed, cloudiness and solar radiation in Manchester, United Kingdom. Ganbat et al. (2013) examined the relative importance of four meteorological parameters for the maximum CLUHII in Ulaanbaatar, Mongolia, including the previous-day maximum CLUHII, wind speed, cloudiness and relative humidity. The percentage of explanation was higher during the daytime (55.6 %) than at nighttime (46.6 %), and obviously lower in the winter (18.5 %) than other three seasons (33.4 %–38.1 %) (Ganbat et al., 2013). The previous-day maximum CLUHII and cloudiness were the two most important parameters (Ganbat et al., 2013).

Some studies reported that heat waves could enhance the CLUHI during the daytime and nighttime (Founda et al., 2015; Ngarambe et al., 2020; Park et al., 2023; Ramamurthy and Sangobanwo, 2016; Rizvi et al., 2019; Schatz and Kucharik, 2015). The CLUHII were 3.30 and 4.50 °C higher during the heat weave than non-heat weave period under the common summer context in Seoul, Korea in 2012 and 2016, respectively (Ngarambe et al., 2020). The synergies were relatively larger in denser built regions and with smaller wind speeds (Ngarambe et al., 2020). The maximum air temperature showed larger significant relationships with CLUHII derived by both maximum and minimum temperature in Seoul, Korea during the the heatwave than non-heatwave period in July and August (Park et al., 2023). Nevertheless, no statistical relationship existed between CLUHII and maximum temperature in Rome, Italy (Cecilia et al., 2023) or mean temperature in Dijon, France (Richard et al., 2021). The CLUHI phenomenon was strong in sunny but not windy weather. Although heat waves often occurred in such conditions, they were not synchronized (Richard et al., 2021). Sunny but not windy conditions could also happen without high temperatures (Richard et al., 2021). Meanwhile, Rogers et al. (2019) found the heatwave exacerbated the diurnal changes of a typical CLUHI, resulting in larger and smaller CLUHII during the night and daytime, respectively. Moreover, the CLUHII diminished evidently during the heatwave in the surrounding Perth (Rogers et al., 2019). In addition, Scott et al. (2018) found the CLUHII diminished with warmer temperatures in 38/54 American cities. The lowest CLUHII nights occurred under moist weather. The warming cities had not experienced an enlarging CLUHII. Peng et al. (2019) found the CLUHII was larger in cities in arid than humid areas, but no obvious differences occurred in cities among different climatic zones in China. Finally, the influence of air temperature on the annual CLUHII was positive, but it was negative in summer in Hong Kong (Zheng et al., 2023a). The relationship between CLUHII and temperature showed obvious diurnal variations on the hourly scale in Rome, Italy (Cecilia et al., 2023).

3.2. *Air pollution*

The CLUHII was positively correlated with carbon monoxide (CO), nitrogen dioxide (NO_2), sulfur dioxide (SO_2) and particulate matter (PM), and negatively correlated with Ozone (O_3) in Seoul, Korea (Ngarambe et al., 2021). The effects of air pollution on CLUHII showed substantial seasonal variations, the largest for NO_2 and PM_{10} in the winter and SO_2, CO, $PM_{2.5}$ and O_3 in the autumn (Ngarambe et al., 2021). The CUHII derived by the daily maximum/minimum temperature was negatively/positively correlated with $PM_{2.5}$, resulting in a decrease in the diurnal temperature range (Yang et al., 2020). Wang et al. (2022) found the impacts of $PM_{2.5}$ on CLUHII could change under different wind directions in Beijing, China. The CLUHII was enhanced by $PM_{2.5}$ under the north wind during the daytime and nighttime, while weakened under the south and west wind. The differences in $PM_{2.5}$ pollution between urban and rural stations weakened the nighttime CLUHII in the summer and winter, but strengthened the daytime CLUHII in the winter in Beijing, China (Yang et al., 2021). The correlation between the CLUHII and $PM_{2.5}$ concentration was regulated by the interaction of aerosol with radiation, evaporation and planetary boundary layer height (Yang et al., 2021). The former two changed the surface energy balance via sensible and latent heat fluxes, while the latter affected atmospheric stability and energy exchange (Yang et al., 2021).

3.3. *Socioeconomic factors*

Oke (1973) found the positive effect of the total population on CLUHI and constructed a regression equation between CLUHII and the logarithm of the total population in Quebec, Canada. Similar findings also applied to cities in North America, Europe and hot and humid tropical regions (Chow and Roth, 2006). Moreover, the CLUHII was significantly positive with the total registered population in Shanghai, China (Zhang et al., 2010) and population density in the Netherlands (Wolters and Brandsma, 2012). A positive logarithmic relationship existed between the CLUHII and population density in Shenzhen, China (Lei et al., 2014).

The CLUHII was significantly positive with the manufactured dwelling house area in Beijing, China (Yu, 2005) and significantly negatively linearly with the cropland area in Shanghai, China (Zhang et al., 2010). Moreover, a positive logarithmic relationship existed between the CLUHII and gross domestic product (GDP), electric power consumption and area of paved roads in Shanghai, China (Zhang et al., 2010). In addition, the Coronavirus Disease 2019 (COVID-19) lockdown decreased CLUHII significantly, especially during the nighttime in Wuhan, China (Sun et al., 2022).

The grey relational grades (GRGs) were all larger than 0.5 between the CLUHII derived by the maximum, mean and minimum temperature and four socioeconomic factors, including the urban resident population, GDP, fixed investments and gross industrial output value in Chongqing, China (Zhang et al., 2022a). The GRG was larger for the urban resident population than for urban economic factors (Zhang et al., 2022a). The tertiary regression relationship was also established between CLUHII and the urbanization index constructed by the four above-mentioned socioeconomic indices using the principal component analysis, indicating that CLUHII would continue to grow with urbanization in Chongqing, China (Zhang et al., 2022a). Peng et al. (2019) found no significant differences existed for CLUHII among cities with different population sizes, GDP levels and industrial structures by analyzing 155 cities in China.

3.4. Urban underlying surface condition

In Singapore, the nighttime CLUHII was the highest on commercial land, lowest on low-rise residential land, and similar in the Central Business District and high-rise residential land (Chow and Roth, 2006). The land cover type for the daytime CLUHII was commercial land > low-rise residential land > Central Business District > high-rise residential land (Chow and Roth, 2006). In general, there were significant differences in CLUHII between the first and last two land use types (Chow and Roth, 2006). Moreover, the CLUHII was smaller for the sites in ten pocket parks than their surrounding urban streets during the daytime and nighttime in Hong Kong, China (Lin et al., 2017).

The daytime CLUHII was significantly positive with the building footprint ratio and asphalt space ratio in Xi'an, China (Xu et al., 2019). A positive logarithmic relationship existed between the CLUHII and road cover percentage in Shenzhen, China (Lei et al., 2014). The road area had positive effects on CLUHII in Tainan, China, especially during the nighttime (Chen et al., 2019). However, higher floor area ratio (FAR) and building density in high-rise high-density urban environments reduced daytime CLUHII without increasing the early nighttime CLUHII in Hong Kong, China (Lin et al., 2017).

The daytime CLUHII were significantly negative with the green land ratio, and tree canopy ratio (TCR) in Xi'an, China (Xu et al., 2019), with the TCR in Hong Kong, China (Lin et al., 2017). The TCR showed little influence when its value was smaller than 42 %, and strongly negative impacts of TCR on CLUHII were found when the TCR value increased from 42 % to 64 % in Hong Kong, China (Lin et al., 2017). Negative impacts of vegetation were found during both the daytime and nighttime in Tainan, China (Chen et al., 2019). The normalized difference vegetation index (NDVI) correlated negatively and positively with CLUHII derived by the minimum and maximum temperature respectively, in Baghdad, Iraq (Al-Saadi et al., 2020). The water area decreased the daytime CLUHII in Tainan (Chen et al., 2019) and Xi'an, China (Xu et al., 2019). Some scholars also mentioned the shadow in urban areas may weaken CLUHII and even lead to the occurrence of cold CLUHI in Salamanca, Spain (Alonso et al., 2003).

3.5. Inland or coastal type and landform condition

The inland cities exhibited higher CLUHII than coastal ones in China (Peng et al., 2019) and Korea (Kim and Baik, 2004). The positive and negative effects existed between CLUHII and the distance to sea during the daytime and nighttime in four seasons in Tainan, China, respectively (Chen et al., 2019). The cities in middle or high mountain areas experienced larger CLUHII than in other landforms in China (Peng et al., 2019).

3.6. Combined effects

Oke (1973) built a regression equation between the logarithm of CLUHII and influence factors, including total population and wind speed in Quebec, Canada. Chen et al. (2019) proposed the regression equation between CLUHII and four influence factors, including the distance to the sea, floor area, road area, vegetation area and water area during the daytime and nighttime in four seasons in Tainan, China. The degree of explanation was larger during the nighttime than daytime, ranging from 30 % to 70 % (Chen et al., 2019).

4. Future directions

4.1. Improvement of data acquisition, sharing and use

Eight types of air temperature data have been used to obtain excellent progress in CLUHI research (Table 1). Some directions of efforts based on the existing issues were as follows. Firstly, the data obtained by the traditional observation method of sparse observation stations can no longer meet the needs of current CLUHI research. High-density meteorological measurement networks should be encouraged to be built in more cities (especially the important or representative ones). Secondly, it is not easy, sometimes very difficult, to obtain the existing meteorological data from either traditional sparse observation sites or later high-density observation sites. Promoting data sharing is urgent and necessary, but hard to achieve and requires a sustained effort from multiple parties, such as the government, scholars, and residents (especially the weather amateurs) (Cecilia et al., 2023; Golroudbary et al., 2018; Wolters and Brandsma, 2012), etc. Thirdly, the hysteresis effect of the instrument probe needs to be considered. Mobile devices need to stay at the measuring point for longer than a certain amount of time to get reliable temperature data, especially in complex urban environments with high heterogeneity and drastic changes in LC/LU and thermal environment. Fourthly, it is highly encouraged to use remote sensing, numerical simulation, data assimilation, down-scaling and other technologies to obtain and share the data sets of air temperature or various apparent temperature indices in higher temporal and spatial resolutions that can cope with various weather conditions. For instance, Zhang et al., (2022b) offered the global daily maximum and minimum temperature at 1 km resolution during 2003–2020. Yao et al. (2023) shared a global monthly 30 arcsecond resolution seamless land surface and near-surface temperature data under clear sky and all-weather conditions from 2001 to 2020. Pede and Mountrakis (2022) modeled the daily maximum heat index at 1 km resolution from May through September of 2012 across the conterminous United States. Zhang et al. (2023) published the monthly human thermal index collection at 1 km resolution over China during 2003–2020. The European Centre for Medium-Range Weather Forecasts (ECMWF) released an hourly enhanced global dataset for the land component of the fifth generation of European ReAnalysis (ERA5), hereafter referred to as ERA5-Land from 1950 to 2022 (Muñoz-Sabater et al., 2021). Fifthly, attention should be paid to the suitability and accuracy of the above data sets in analyzing CLUHI. It is necessary to analyze not only the precision of air temperature and apparent temperature indices of the above-mentioned various data sets in different climate zones, landforms, LC/LU, weather conditions, etc. but also the differences in CLUHII derived by using such data sets and the traditional observed data in the measuring points. Finally, although many urban thermal environment studies have been carried out using numerical models such as WRF or CFD, their applicability in different climatic zones, urban contexts and landforms should be further verified. In addition, it is also worth discussing how to use a numerical model to analyze the urban thermal environment under non-clear sky conditions because such days often occur in many cities in the hot humid period.

4.2. Considering apparent thermal indices

Human thermal comfort is not only related to the air temperature but also affected by the relative humidity, wind speed, solar radiation, etc., especially under the hot humid and cold windy conditions (Zhang et al., 2023). It is urgent and necessary to study CLUHI using the thermal comfort indices derived from several meteorological factors. Such studies were sorely lacking (see section 2.2 for the details). Other related studies were mainly as follows. Pecelj et al. (2021) comparatively analyzed the differences between the meteorological stations in the urban centre park and airport region at 7:00 and 14:00 for PET, modified PET and universal thermal climate index. Yao et al. (2022) found the temperature in urban areas was higher than that in rural areas in the

Yangtze River Delta urban agglomeration in China. Nevertheless, the intensity and duration of heat waves in urban areas were lower than those in surrounding areas because of the higher relative humidity in rural areas. Moreover, this region had a higher probability of extreme temperatures during 1991–2019 but had a decreasing trend of heat weave due to the decline of relative humidity. In addition, some studies analyzed the variation of thermal comfort indices within cities and associated influence factors (Van de Walle et al., 2022; Xu et al., 2019).

4.3. Focus on the rarely studied regions, cities, and scales

The existing research mainly focused on Europe, North America and East Asia. Other parts of the world had limited or even no research, especially in the underdeveloped, low latitudes and elevations areas with high population density or urbanization rate, or rapid growth of population or urbanization rate. For one thing, the heat stress is severe in these places. On the other hand, several factors have promoted the thermal environment in those areas likely to deteriorate. The corresponding negative effects would be greater, including decreasing thermal comfort, affecting work efficiency, damaging health, increasing morbidity and mortality, enlarging energy consumption of refrigeration, magnified emissions of greenhouse gas and pollution, etc. The factors that led to the aggravation of heat stress mainly included global warming, population growth, urbanization promotion, economic growth, improvement of people's living standards, etc. For instance, the increased temperatures in the hot regions could easily amplify the effects of heat stress due to the high background air temperature. On the contrary, the enlarged temperature in regions with low background temperature had a small or no negative effect on heat stress and even was beneficial for thermal comfort there. These inferences can be supported by the finding that the increasing rates of cooling degree days have become larger at lower latitudes and attitudes (Li et al., 2023a). These sensitive and fragile regions mainly included South Asia, Southeast Asia, West Asia, the southern half of West Africa, Central Africa, the northern marginal areas of South America, the southern coastal areas of China, etc. (Li et al., 2023a).

Most previous studies analyzed the CLUHII in large cities, but 59 % of the world's urban residents lived in small and medium-sized cities with less than 1 million population (DESAPD, 2018). Meanwhile, the CLUHI could be observed in a small town with just 1000 people (Oke, 1973). Several studies have been carried out on the SUHI in cities of various sizes (including small cities) (Clinton and Gong, 2013; Li et al., 2022), which could provide some reference for CLUHI studies (Fig. 5).

Moreover, most of the papers were carried out at the urban scale, while a small number of representative studies on the regional scale, including analyzing CLUHI in all cities in the Netherlands (Golroudbary et al., 2018; Wolters and Brandsma, 2012), six cities in Korea (Kim and Baik, 2004), 86 (Zhou et al., 2023) and 200 cities in China (Peng et al., 2019), and 10 (Ramamurthy and Sangobanwo, 2016), 50 (Stone Jr, 2007) and 54 cities in the United States (Scott et al., 2018), etc. Large-scale studies of CLUHI are necessary and urgent, including regional, national, continental and even global scales. Several studies on SUHI have been done on a large (even global) scale (Clinton and Gong, 2013; Peng et al., 2012; Yao et al., 2019). Only on large scales could we accurately and comprehensively grasp the spatiotemporal variation of CLUHI and its influencing factors, and better guide urban planning and coping with climate change.

4.4. Improving the calculation of canopy urban heat island intensity

Urban-rural comparison method was a traditional commonly used method to calculate CLUHII. Identifying the urban and rural stations was the most basic and initial work. Nevertheless, there was some uncertainty, especially when using the population as an indicator collected based on the administrative units (please see details in section 2.3). The remote sensing method should be recommended to classify stations because it can accurately obtain the background details around the measuring points. The LC/LU is a better indicator than NLI and LST, especially for studies in large regions due to its higher spatial resolution, longer period, more stable threshold, increasing number of LC/LU data sets, etc. For instance, Zhang et al. (2021) published the global land cover data at 30 m resolution from 1985 to 2020 at five-year intervals.

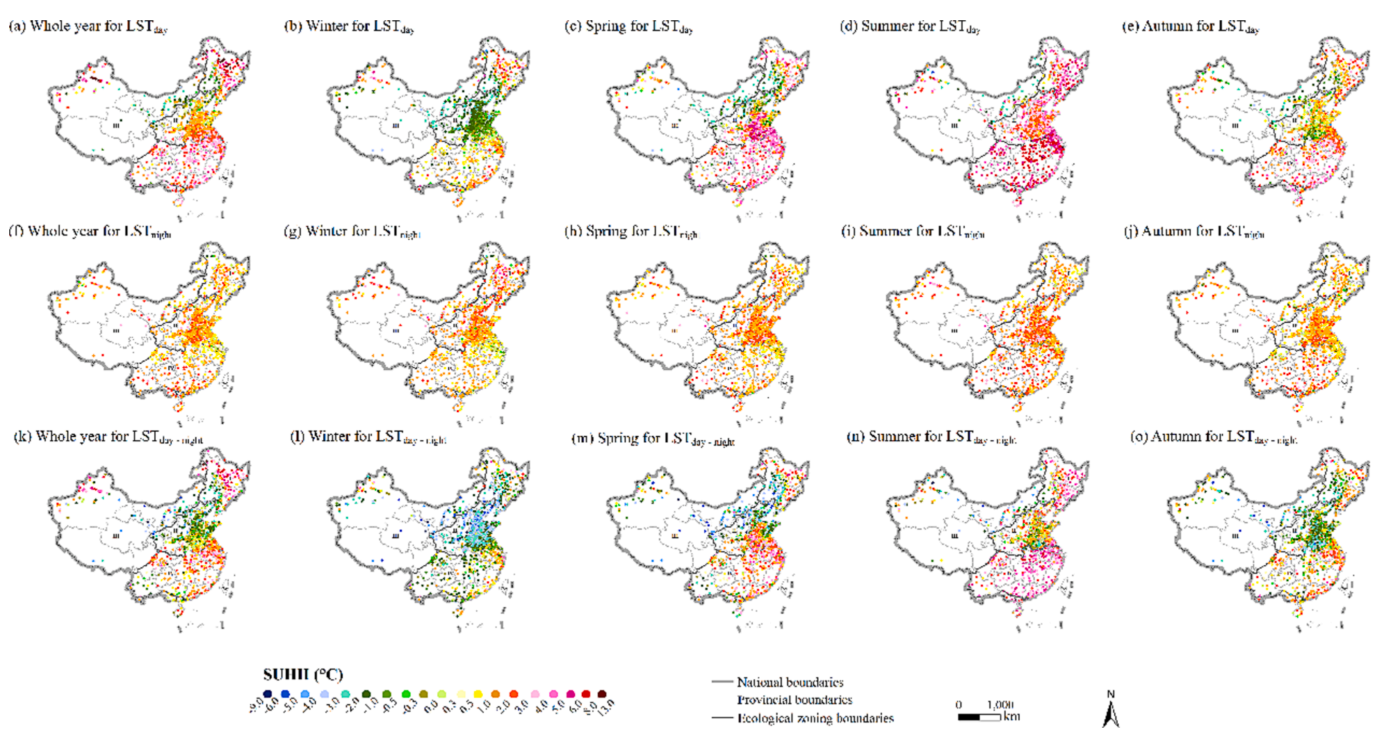

Fig. 5. Annual and seasonal surface urban heat island intensity (℃) at about 13:30 and 01:30 in 917 urban region agglomerations in five ecological zones of China during 2015–2019. The land surface temperature data were derived by Aqua/MODIS. The definition of urban and rural areas has been referenced by Li et al. (2022).

Jun et al. (2014) shared the global land cover data at 30 m resolution in 2000, 2010 and 2020. Huang et al. (2021) produced global impervious surface data at 30 m resolution from 1972 to 2019. The Environmental Systems Research Institute provided the global Sentinel-2 LC/LU time series at 10 m resolution from 2017 to 2021. Google published the global land use data at 10 m resolution in near real-time (Brown et al., 2022). Li et al., (2023b) offered the first national-scale land-cover map at 1 m resolution in China. In addition, referring to the research of SUHI (Zhou et al., 2014), the method of dividing urban and rural sites or regions should be highly recommended by the land use, elevation and distance from urban areas (Sun et al., 2022), especially when using the near-surface air temperature or apparent temperature data obtained using remote sensing (please see details in section 2.1.2).

The CLUHII and its variation pattern could vary in the same city if different reference stations were selected (Golroudbary et al., 2018; Rizvi et al., 2019; Schlünzen et al., 2010; Wolters and Brandsma, 2012). Therefore, it should be encouraged to use the high-density site data to select multiple urban and rural sites for a city, aiming to accurately explain its spatiotemporal variation of CLUHII. In addition, due to the limited number of traditional weather stations, it was sometimes difficult to find qualified rural stations, especially in densely populated or economically developed areas (Li et al., 2019). Therefore, it was highly recommended to calculate CLUHII by constructing the correlation between PIS and air temperature. Note there was also a similar study on

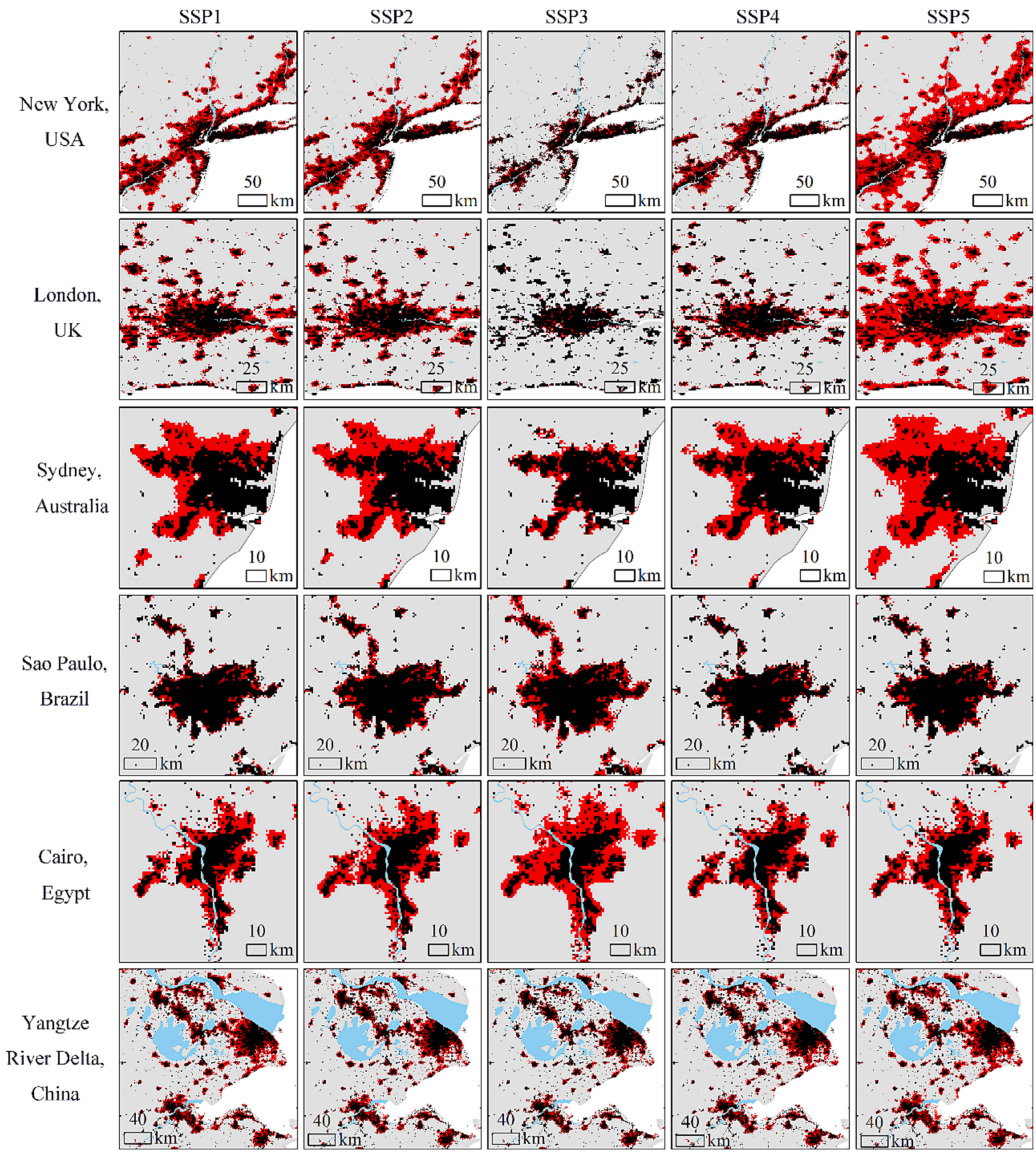

Fig. 6. Projected urban land by 2100 in global six metropolitan areas under five shared socioeconomic pathways (SSPs). Data source: Geographical Simulation and Optimization Systems (http://www.geosimulation.cn/GlobalSSPsUrbanProduct.html).

the calculation of SUHII (Li et al., 2018b). However, this method could not be suitable in too large regions, because the temperature could be affected by other more important factors, such as latitude.

4.5. Attention to more associate factors and driving force analysis methods

More influence factors should be considered or further analyzed, including the cold wave, wind direction, fog, building footprint ratio, floor area ratio, building height, volume and surface area, sky view factor, urban shape index, distance to large waterbodies, differences between the urban and rural stations or regions in wind speed, relative humidity, air pollution, population density, GDP per unit of land area, NLI, albedo, PIS, green land ratio, TCR, waterbody ratio, urban shadow ratio, remotely sensed indices of building, vegetation, water or bare soil, various landscape indices, etc. The release of relevant data sets could provide convenience for such research, including the above-mentioned various LC/LU data (please see details in section 4.4), global 777 million building footprints in recent years by Bing Maps (Bide et al., 2023), global gross domestic product and electricity consumption at 1 km resolution from 1992 to 2019 (Chen et al., 2022), the global population at 1 km resolution during 2000–2020 (Lloyd et al., 2017), building height at 10 m resolution in 2020 in China (Wu et al., 2023), Luojia1-01 NLI data at 130 m resolution in China (Guo et al., 2023), daily coupled evapotranspiration data at 500 m resolution in China in the recent 20 years (He et al., 2022), daily $PM_{2.5}$ at 1 km resolution in recent two decades in China (Wei et al., 2021), daily $PM_{2.5}$, SO_2 and O_3 at 1 km resolution during 2015–2020 in China (Chi et al., 2023), etc. In addition, some associated factors could be obtained in the future. For instance, the global urban land expansion was simulated for the 21st century under the shared socioeconomic pathways at 0.125°(Gao and O'Neill, 2020) and 1 km resolution (Fig. 6) (Chen et al., 2020), respectively. Zhuang et al. (2022) projected urban land expansion at 30 m resolution through 2050 in China under the shared socioeconomic pathways. Olén and Lehsten (2022) published the global annual population distribution at 1 km resolution developed with the CMIP6 RCP and SSP scenarios during 2010–2100.

The method in previous studies to analyze the associated factors of CLUHII, including the comparative analysis (Acero et al., 2013; Bokwa et al., 2018; Ramamurthy and Sangobanwo, 2016), bivariate correlation analysis (Al-Saadi et al., 2020; Richard et al., 2021; Schatz and Kucharik, 2015), regression analysis (Levermore et al., 2018; Ngarambe et al., 2021; Schatz and Kucharik, 2015), etc. More methods could be encouraged to be used, such as the partial correlation analysis (Harmay and Choi, 2023; Li et al., 2022; Zheng et al., 2023b), relative importance analysis (Li et al., 2020; Li et al., 2023a), geographic detector (Yan et al., 2022; You et al., 2021), grey relational analysis (Yan et al., 2022; Yi et al., 2021), machine learning method, such as neural network model (Kim and Baik, 2002; Li et al., 2023a; Li et al., 2019), random forest method (Guo et al., 2023; Ho et al., 2016; Pede and Mountrakis, 2022), Cubist model (Rao et al., 2019; Yao et al., 2021), light Gradient Boosting Machine algorithm (Cho et al., 2022; Zhang et al., 2023), etc. In addition, the driving mechanisms of CLUHII vary during the daytime and nighttime, among different seasons and regions, and should be treated separately rather than mixed.

5. Conclusions

This paper aimed to review the rich achievements in CLUHI research from the aspects of monitoring and associated factors. The main findings were as follows.

(1) Three phases can be divided for this research direction, including the initial stage (1967–1990), the slow acceleration period (1991–2010) and the accelerated development phase (2011–now). Since 2000, the number of related articles has shown a "J-shaped" growth.

(2) Eight methods have been mainly adopted to obtain the surface air temperature data for the CLUHI study, including four observation methods (Meteorological station observation, high-density meteorological measurement network, in-situ fixed-points observation and moving transect observation), two numerical modeling methods (Urban boundary layer model coupled with the urban canopy model, and computational fluid dynamics model), remote sensing and data assimilation methods.

(3) Much previous research used near-surface air temperature to study CLUHI. Various apparent temperature indices have been proposed, which could indicate the thermal comfort condition more accurately than temperature. Obvious differences existed between the air temperature and apparent temperature, especially under humid hot or cold windy conditions. Nevertheless, rare studies analyzed CLUHI using the apparent temperature indices.

(4) The CLUHII was generally defined as the differences in near-surface temperature between the urban and suburb or rural stations or regions, derived by the subjective experience, population, LC/LU, LCZ, NLI, LST, spatial location, etc. Some scholars avoided disputes over the identification of urban and rural areas and instead defined the CLUHII as the temperature differences between regions with 0 % and 100 % PIS, based on the regression analysis between temperature and PIS.

(5) The diurnal, monthly, seasonal or interannual change of CLUHI has been generally monitored in various global regions, especially Europe, North America and East Asia.

(6) Six types of associated factors of CLUHI have been analyzed, including the meteorological condition, air pollution, socioeconomic factors, urban underlying surface condition, inland or coastal type and landform condition, and the combined effects of several influence factors. The most widely used methods included comparative analysis, bivariate correlation analysis, regression analysis, etc.

(7) Five potential research directions in the future were proposed, including the improvement of data acquisition, sharing and use, use of apparent thermal indices, focus on the rarely studied regions, cities, and scales, improving the calculation of CLUHII, and attention to more associate factors and driving force analysis methods.

CRediT authorship contribution statement

Yuanzheng Li: Conceptualization, Data curation, Funding acquisition, Investigation, Methodology, Project administration, Writing – original draft, Writing – review & editing. **Tengbo Yang:** Data curation, Validation, Visualization, Writing – review & editing. **Guosong Zhao:** Investigation, Writing – review & editing, Conceptualization. **Chaoqun Ma:** Data curation, Formal analysis, Visualization. **Yan Yan:** Visualization, Writing – review & editing, Funding acquisition. **Yanan Xu:** Project administration, Validation. **Liangliang Wang:** Investigation, Methodology. **Lan Wang:** Conceptualization, Funding acquisition, Project administration, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data generated or analyzed during this study are available from the corresponding author upon reasonable request.

Acknowledgements

This research was funded by the Key Scientific Research Project of Henan Colleges and Universities (Grant No. 24A170005 & 22A170003), Natural Science Foundation of China (Grant No. 31600375 &

41901238), Henan Provincial Youth Natural Science Foundation (Grant No. 222300420104), Henan Philosophy and Social Science Planning Project (Grant No. 2021BJJ002), Key Scientific and Technological Research Projects of Henan Province (Grant No. 212102310005).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ecolind.2023.111424.

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